EE619 VLSI System Design 2025-26-II

Assignment 2 **Dated: 16/02/2026** **Due Date : 8/03/2026**

Instructions:

1. Please use Virtuoso with gpdk180 transistor models. Refer to RV_CMOS_INV_Schematic_design.mp4 to learn how to do this. The path to be added to the library path editor is /home/vlsi_IITK/GPDK/gpdk180_v3.3/libs.oa22/gpdk180.
2. Please do the tool setup similarly as in Assignment-1 using materials on Mookit.

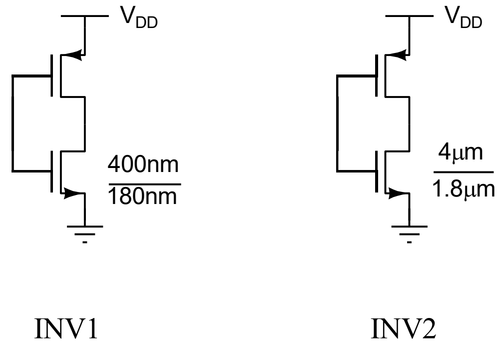

Figure 1: Schematic for INV1 and INV2.

Q1. Consider INV1, an inverter shown in Fig. 1. The size for the NMOS transistor is shown in the figure.

(a) Size the PMOS transistor in INV1 to obtain VTC with switching threshold $\sim V_{DD}/2$, where V_{DD} = 1.8V. Please do this for the process corner nn, at temperature 27°C. Your transistor could come in nominal, fast or slow corner. Please refer to the lecture video by clicking here (https://www.youtube.com/watch?v=6bKcCrAildU&list=PLP-rjhz_nIi4kBIvyb6dN6OmYSNSWJioV&index=12) from 6mins-21mins to learn about this.

To select transistor properties to a particular corner, you have to select the appropriate section in the transistor model file. In the ADE-L window, click on Setup→Model Libraries. The Model Library Setup window will open, as shown in Fig. 2 (a). Double click on the section associated with the model file to get a drop-down window. You can select the appropriate corners. nn stands for nominal nominal, indicating that both the NMOS and PMOS transistors are in the nominal corner. Temperature is directly displayed in the ADEL window as shown in Fig. 2(b).

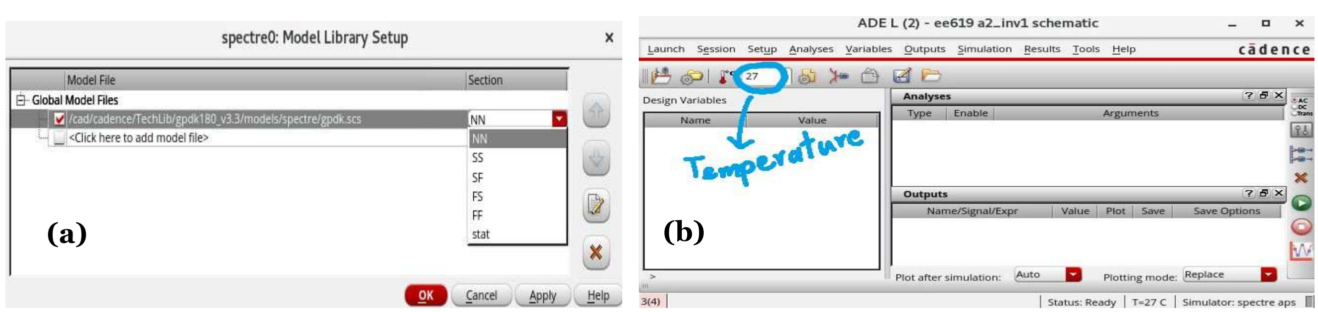

Figure 2: (a) Model Library Setup window, (b) ADEL window where the temperature option is highlighted.

(b) Simulate and plot the VTC for INV1 at 27°C for the following corners.

- nn
- ss
- ff
- sf
- fs

(c) Estimate the signaling thresholds, V_{IH}, V_{IL}, V_{OH} and V_{OL} for this inverter based on the VTCs obtained in 1.(b).

Q2. Consider INV2, an inverter shown in Fig. 1. The size for the NMOS transistor is shown in the figure.

(a) Size the PMOS transistor in INV2 to obtain VTC with switching threshold $\sim V_{DD}/2$, where $V_{DD} = 1.8$ V is the supply voltage. Please do this for the process corner nn, at temperature 27°C.

(b) Simulate and plot the VTC for INV2 at 27°C for the following corners.

- nn
- ss
- ff
- sf
- fs

(c) Estimate the signaling thresholds, V_{IH}, V_{IL}, V_{OH} and V_{OL} for this inverter based on the VTCs obtained in 2.(b).

Q3. Plot the VTCs obtained for INV1 and INV2 at nn 27°C in the same graph and compare the curves. What differences do you observe?

Q4. A designer is building a circuit using INV1 and INV2. Based on (1.c) and (2.c), determine the signaling thresholds, noise margins and the noise immunity for the design.

Q5. Simulate the propagation delays (t_{pLH}, t_{pHL}) of INV1 at nn, 27°C corner under the following conditions.

(a) Simulate the propagation delay for load capacitance, $C_L = 0.1$ pF, when the input rise/fall time is 10 ps, 50 ps, 100 ps and 500 ps. Give these rise/fall times as properties in the vpulse source.

(b) Simulate the propagation delay for input rise/fall time of 50 ps, when the load capacitance is 0.05 pF, 0.1 pF and 0.5 pF.

(c) Assume that an inverter drives $C_L = 0.1$ pF. The rise/fall time of the input is expected to be within 10 ps to 50 ps. The designer wants to ensure that the propagation delays are less than 2 ps. Is this achievable by sizing the transistors in INV1? If so, how will you proceed with the design? If not, what is limiting you? What is the minimum possible delay you can achieve?

Q6. Simulate the propagation delays of INV2 at nn, 27°C for load capacitance $C_L = 0.1$ pF and input rise/fall time = 50 ps. How does this compare to the propagation delays in INV1 under the same conditions?

Q7. Assume that INV1 is driving $C_L = 0.05$ pF. The input to the inverter is a square wave clock signal with rise/fall time = 500 ps. Simulate the below cases for nn, 27°C.

(a) Assume that the clock frequency is 100 MHz. Plot the current drawn by INV1 from the supply for simulation time from 0 ns to 50 ns. Observe the waveform.

(b) Estimate the power consumption of INV1 for the situation described in 7.(a). Is it higher or lower than the theoretical estimate? Why?

(c) The clock frequency is increased to 200 MHz. Estimate the power consumption in INV1 now. Compare this power with the power obtained in 7.(b).

(d) Assume that the clock frequency is 100 MHz, with rise/fall time increased to 1 ns. Load capacitance is the same. Estimate the power consumption in INV1. Compare this power with the power obtained in 7.(b).

Submission Guidelines:

File Name: RollNo_Assignment2_EE619.pdf

Question 1

Fig-1: Schematic of INV-1 with the sizes of NMOS and PMOS as described.

Fig-2: Plot the VTC for the five corners as explained in question (1.b). This should be a single plot, with five curves. Clearly mark the V_{IH}, V_{IL}, V_{OH} and V_{OL} for INV-1 in the plot.

Explanation-1: Explain briefly how the signaling thresholds for INV-1 were estimated. You may use figures for the explanation, if necessary.

Question 2

Fig-3: Schematic of INV-2 with the sizes of NMOS and PMOS shown.

Fig-4: Plot the VTC for the five corners as explained in question (2.b). This should be a single plot, with five curves. Clearly mark the V_{IH}, V_{IL}, V_{OH} and V_{OL} for INV-2 in the plot.

Explanation-2: Explain briefly how the signaling thresholds for INV-2 were estimated. You may use figures for the explanation, if necessary.

Question 3

Fig-5: Plot the VTCs as described in question-3.

Explanation-3: What are your observations after comparing the VTCs of both inverters?

Question 4

Table-1: Tabulate the V_{IH}, V_{IL}, V_{OH}, V_{OL}, noise margins and the noise immunity for the design described in question-4.

Explanation-4: Explain briefly how the values in table-1 were estimated.

Question 5

Table-2: Tabulate t_{pLH} and t_{pHL} for INV-1 for different rise/fall times and for different loads.

Explanation-5: Your comments for question (5.c).

Question 6

Table-3: Tabulate t_{pLH} and t_{pHL} for INV-1 and INV-2 for the condition described in question-6. Explanation-6: Your comments for question-6.

Question 7

Fig-6: Plot the current waveform as described in question (7.a).

Explanation-7: Estimate the power consumption. Explain briefly how the power consumption was estimated and add your comments for question (7.b).

Explanataion-8: Estimate the power consumption. Your comments for question (7.c).

Explanation-9: Estimate power consumption. Your comments for question (7.d).